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 Studierichting: Informatica
 Oefeningenreeks 1D nr. 6.9 [1]

$$((2+i)z^3 - iz^2 + 3) \div ((1-i)z^2 - z + 1)$$

$(2+i)z^3 \quad -iz^2 \quad +0z \quad +3$	$(1-i)z^2 - z + 1$
$- \quad (2+i)z^3 \quad -\frac{1+3i}{2}z^2 \quad +\frac{1+3i}{2}z \quad $	$\frac{1+3i}{2}z + \frac{1}{2}i \rightarrow Q$
$0 \quad +\frac{1+i}{2}z^2 \quad -\frac{1+3i}{2}z \quad +3$	
$- \quad \frac{1+i}{2}z^2 \quad -\frac{1}{2}iz \quad +\frac{1}{2}i$	
$R \leftarrow \left(-\frac{1}{2} - i\right)z \quad +3 - \frac{1}{2}i$	

Resultaat:

$$A(z) = B(z) \cdot Q(z) + R(z)$$

$$A(z) = ((1-i)z^2 - z + 1) \cdot \left(\frac{1+3i}{2}z + \frac{1}{2}i\right) + \left(\left(-\frac{1}{2} - i\right)z + 3 - \frac{1}{2}i\right)$$

KladBlok:

Berekening van het eerste quotiëntterm:

$$\begin{aligned}
 \frac{(2+i)z^3}{(1-i)z^2} &= \frac{(2+i)}{(1-i)}z \\
 &= (2+i) \cdot \frac{(1+i)}{1^2+1^2}z \\
 &= \frac{2+2i+i+i^2}{2}z \\
 &= \frac{1+3i}{2}z
 \end{aligned}$$

Berekening van het 2de quotiëntterm:

$$\begin{aligned}
\frac{\frac{1+i}{2}z^2}{(1-i)z^2} &= \frac{1+i}{2(1-i)} \\
&= \frac{(1+i)(1+i)}{2(1^2+1^2)} \\
&= \frac{1+2i+i^2}{4} \\
&= \frac{2}{4}i = \frac{1}{2}i
\end{aligned}$$

References

[1]